AAREEN SATISH AHER

AI/ML Engineer

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SUMMARY

Apply machine learning and AI systems to solve real-world problems. Experienced in building predictive models, developing end-to-end ML pipelines, and deploying scalable AI solutions using Python and SQL to generate impactful insights. Skilled in transforming complex datasets into actionable insights that support data-driven decision making.

EDUCATION

Dual Degree (B.Tech. + M.Tech.) Environmental Science & Engineering

Indian Institute of Technology, Bombay

11/2020 - 05/2025 | Mumbai, India

GPA **3.2** / 4.0

EXPERIENCE

Machine Learning Researcher (Master's Thesis)

ANT Lab, IIT Bombay

03/2024 - 05/2025 | Mumbai, India

- Developed an end-to-end PM2.5/PM10 time-series forecasting pipeline across 3 stations for 72 hour prediction.
- Trained baseline models using XGBoost, ANN, and LSTM achieving R^2 up to 0.94 under out-of-sample evaluation.
- Engineered five hybrid ensemble methods reducing forecast RMSE by 18–30% and compressing error variance by 35%.
- Applied Bayesian Model Averaging achieving PM2.5 RMSE of 1.43 μg/m³ and PM10 RMSE of 4.71 μg/m³ across Mumbai.

Data Analysis Intern

University Of Alberta

05/2023 - 07/2023 | Edmonton, Canada

- Optimized road-geometry impact assessment workflows by analyzing traffic data rates using the VISTA simulator
- Increased data extraction accuracy by 15% by developing a pipeline that isolates pavement from 500+ road sections
- Analyzed LIDAR point cloud data from over 100+ scans with Cloud Compare for reliable surface-level insights

SKILLS

Machine Learning & AI

Predictive Modeling, Random Forest, XGBoost, ANN, LSTM, GenAI, DenseNet121, Explainable AI, PDP, ALE, RAG, SHAP, Large Language Models, OpenCV, Semantic Search, Prompt Engineering, Natural Language Processing, CNN

MLOps

LangChain, ChromaDB, DagsHub, Vector Databases, OpenAI Embeddings, Streamlit, Flask, Docker, n8n, AWS (EC2, ECR), CI/CD (GitHub Action), MLflow, DVC, Experiment Tracking, Git, GitHub, Model Deployment

Data Science & Analytics

Python, SQL, Pandas, NumPy, Scikit-learn, Exploratory Data Analysis, Data Visualization, Power BI, Keras, PyTorch, ETL Pipelines, Seaborn, Microsoft SQL Server, Matplotlib, Data Preprocessing, Data Mining, Statistical Analysis, Business Analytics, PostgreSQL, TensorFlow

PROJECTS

Chest CT Diagnosis System

12/2025 - 01/2026

- Integrated a 2-stage DenseNet121 pipeline on 500+ CT scan images, 99.6% accuracy (binary) and 71.6% multi-class accuracy across 3 carcinoma subtypes.
- Established a reproducible MLOps workflow using DVC and MLflow (DagsHub) for pipeline versioning and experiment tracking across 3 model variants.
- Operationalized a Dockerized Flask service on AWS EC2 via ECR and GitHub Actions CI/CD, delivering real-time CT inference with Grad-CAM explainability.

Multimodal RAG PDF Assistant

11/2025 - 01/2026

- Constructed multimodal RAG pipeline using ChromaDB LangChain, Unstructured, and ChromaDB, processing 10 academic research PDF with title-aware chunking.
- Enhanced semantic retrieval by embedding 1,000+ documents with OpenAI via MMR, improving top-k retrieve relevance by 30% and reducing hallucination using GPT-4o evidence-grounded prompting.
- Launched a streamlit based assistant interface with persistent vector storage, achieving <1.5s average query latency for document question answering.

XAI-Constrained ML Optimization

11/2025 - 01/2026

- Created an explainability-constrained ML pipeline for credit risk modeling across 30+ features using RF.
- Refined models via XAI-guided tuning with SHAP, PDP, and ALE analysis, achieving 93.5% accuracy, 0.95 ROC-AUC, 0.83 F1 and SHAP variance of 2×10^{-3}.
- Automated experiment tracking with MLflow and DagsHub, logging parameters, evaluation metrics, and explainability artifacts for full auditability.

Corporate AI Assistant

11/2025 - 01/2026

- Engineered an enterprise-grade multi-user RAG platform using n8n orchestration and OpenAI embeddings, indexing 10 policy documents.
- Designed LLM-based intent classification pipeline, confidence scoring, and rule-driven access control for governed usage across HR/IT query resolution.
- Prevented unauthorized data access while achieving 0.89 F1-score across internal QA benchmark using explainability-aware confidence thresholds.

Predictive Churn Analytics

11/2025 - 02/2026

- Increased retention by 25% by building an end-to-end churn prediction pipeline using SQL Server ETL and XGBoost, identifying high-risk customers for targeted retention strategies.
- Structured staging and production workflows with validation logic and SQL views, ensuring high data quality, consistency and pipeline reliability.
- Created Power BI dashboard for account churn insights. Aided in data-driven retention decisions.